

# MOHAMMED LOKHANDWALA

Surat, Gujarat

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## EDUCATION

**P.P. Savani University**

*B.Tech in Computer Science (AI & ML Specialization) | CGPA: 8.1*

**Sep 2022 – May 2026**

*Surat, Gujarat*

## EXPERIENCE

**Avinashi Group of Companies**

**Jul 2025 – Nov 2025**

*AI Engineer Intern*

*Surat, Gujarat*

- Engineered a **sub-250ms** bidirectional telephony AI agent for live PSTN calls, streaming audio through STT → LLM → TTS with 20ms micro-batches, real-time noise handling, and barge-in interruption support.
- Built an automated **financial document intelligence** pipeline for scanned statements, invoices, and receipts, combining layout/table detection, OCR, and custom pre/post-processing to produce structured data.
- Built an **agentic video-generation workflow** with a coordinator model orchestrating image, voice, lip-sync, and video-composition models to turn natural-language briefs into multi-scene product ads with validation, retries, and provider fallbacks.
- Developed an **AI interview system** for conversational interviewing and evaluation, extending the internship's work across interactive language-based workflows.
- Designed shared AI infrastructure using **Supabase and a custom API gateway**, with authentication, organization-level access control, quotas, storage, and reusable endpoints for multiple internal AI products.

**Seepossible Innovative Solutions LLP**

**Jun 2024**

*AI/ML Intern*

*Surat, Gujarat*

- Built an internal **image upscaling API** with authentication and configurable model, resolution, and image-size parameters for use as a reusable AI microservice.
- Adapted a **frame-interpolation workflow** for jewelry rendering, allowing roughly half the frames to be rendered directly and reducing rendering and post-production time by about **50%**.

## SELECTED PROJECTS

**Generalized RL Locomotion Platform** | *Python, PyTorch, PPO/SAC, Box2D, Gymnasium, DearPyGui*

- Built a modular 2D reinforcement-learning environment where users define articulated creatures with custom bodies, joints, limits, sensors, and colliders, then train them across procedural terrain.
- Implemented the physics/control loop, LiDAR-style observations, dense reward shaping, parallel training, checkpoint management, and an interactive rigging/evaluation GUI supporting PPO and SAC.

**Local ML Experiment Tracker** | *Python, Node.js, React, TypeScript, SQLite*

- Built an offline ML experiment tracker for logging runs, metrics, configurations, and artifacts, with a Python SDK, CLI, and web dashboard.
- Implemented experiment comparison, metric aggregation, local storage, hardware telemetry, and integrations for common ML training workflows.

**Argus OS** | *Python, LLM Agents, RAG, Custom Memory, Docker*

- Built a multi-agent assistant that searches the **web and email** and ingests local documents to complete multi-step information tasks through shared tools and retrieval.
- Designed a custom long-term memory layer for retaining **facts, decisions, preferences, and task context** across sessions, separate from document retrieval and current task state.
- Added structured tool interfaces and verification flows so the system gathers evidence across sources, answers with references, replans when needed, and requires approval before external actions.

## TECHNICAL SKILLS

**Languages:** Python, C++, JavaScript/TypeScript, SQL

**AI/ML:** PyTorch, Computer Vision, NLP, Transformers, RAG, LLMs, AI Agents, Reinforcement Learning

**Backend/Systems:** FastAPI, Node.js, WebSockets, MongoDB, SQLite, Docker, Linux

**Tools:** Git, GitHub, Weights & Biases

## ACHIEVEMENTS

- **GATE** — cleared in the 3rd and 4th year of B.Tech.
- **SSIP** — awarded ₹30,000 for an AI/robotics innovation project;
- **Hackathon** — **2nd place** in a college hackathon.
- **Open Source** — contributed a scraping-method fix to LNReader; merged upstream.